JAVA SALARY DETAILS PROGRAM

Object-Oriented Programming Practical Record – Program 8

1. Aim

To write a Java program using an interface to calculate an employee's net salary from basic salary, bonus and deduction.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.

2. Create a Java file named **Program8.java**.

3. Create the **SalaryDetails** interface with the calculateSalary() method.

4. Create the **Employee** class and implement SalaryDetails.

5. Initialize basic salary, bonus and deduction values.

6. Implement calculateSalary() in the Employee class.

7. Calculate net salary using basic salary + bonus - deduction.

8. Create an Employee object in main() and call calculateSalary().

3. Algorithm

1. Start the program.

2. Create the SalaryDetails interface with calculateSalary().

3. Create the Employee class that implements SalaryDetails.

4. Set basicSalary to 30000, bonus to 5000 and deduction to 2000.

5. Implement calculateSalary() in Employee.

6. Calculate total salary as basicSalary + bonus - deduction.

7. Display basic salary, bonus, deduction and net salary.

8. Create an Employee object in main().

9. Call calculateSalary() and stop the program.

4. Explanation of the Program

SalaryDetails interface: It declares the calculateSalary() method that must be implemented by the Employee class.

Employee class: It implements SalaryDetails and provides the implementation of calculateSalary().

Salary calculation: Net salary is calculated using basic salary + bonus - deduction.

Employee object: The main() method creates an Employee object and calls calculateSalary() to display the salary details.

5. UML Class Diagram

```
+----------------------------+
|      SalaryDetails         |
+----------------------------+
| + calculateSalary() : void  |
+-------------+--------------+
              |
          implements
              |
      +-------v-------+
      |   Employee    |
      +---------------+
      | basicSalary    |
      | bonus          |
      | deduction      |
      +---------------+
      | calculateSalary|
      +---------------+
```

6. Java Source Code

```
interface SalaryDetails {

    void calculateSalary();
}

class Employee implements SalaryDetails {

    double basicSalary = 30000;
    double bonus = 5000;
    double deduction = 2000;

    public void calculateSalary() {

        double totalSalary =
                basicSalary + bonus - deduction;

        System.out.println("Basic Salary = " + basicSalary);
        System.out.println("Bonus = " + bonus);
        System.out.println("Deduction = " + deduction);
        System.out.println("Net Salary = " + totalSalary);
    }
}

public class Program8 {

    public static void main(String[] args) {

        Employee emp = new Employee();

        emp.calculateSalary();
    }
}
```

6. Java Source Code (continued)

```
interface SalaryDetails {

    void calculateSalary();
}

class Employee implements SalaryDetails {

    double basicSalary = 30000;
    double bonus = 5000;
    double deduction = 2000;

    public void calculateSalary() {

        double totalSalary =
                basicSalary + bonus - deduction;

        System.out.println("Basic Salary = " + basicSalary);
        System.out.println("Bonus = " + bonus);
        System.out.println("Deduction = " + deduction);
        System.out.println("Net Salary = " + totalSalary);
    }
}

public class Program8 {

    public static void main(String[] args) {

        Employee emp = new Employee();

        emp.calculateSalary();
    }
}
```

7. Output of the Program

```
Basic Salary = 30000.0
Bonus = 5000.0
Deduction = 2000.0
Net Salary = 33000.0
```

8. Result

Thus, the Java program successfully demonstrates the use of an interface by implementing SalaryDetails in the Employee class and calculating the employee's net salary.

Student Signature

Faculty Signature